

V4.2 Toy Model 1D - Static Central Source

Relational record density $\rightarrow$ operational cost $\rightarrow$ shortest-path distortion

Setup:

21 x 21 grid; 8-neighbour connections; central source at (0,0).

$R_i = R_0 \exp(-r_i^2 / 2\rho^2)$, with $R_0=1$ and $\rho=2.5$.

$c_{ij} = c_{0,ij} * [1 + \lambda C(a)(R_i + R_j)/2]$.

For this first static run: $C(a)=1$ and $\lambda = 0, 1, 5, 10$.

What the plots show:

Background shading = source field R_i .

Thin paths = bundle from left edge to right edge.

Heavier labelled path = scout ray from (-10,0) to (10,0).

Circle = source spread radius ρ .

Guardrail:

This is not gravitational lensing yet. It is a graph-based operational analogue.

It tests whether local relational density can create delay, detour, path splitting, or focusing.

Numerical summary:

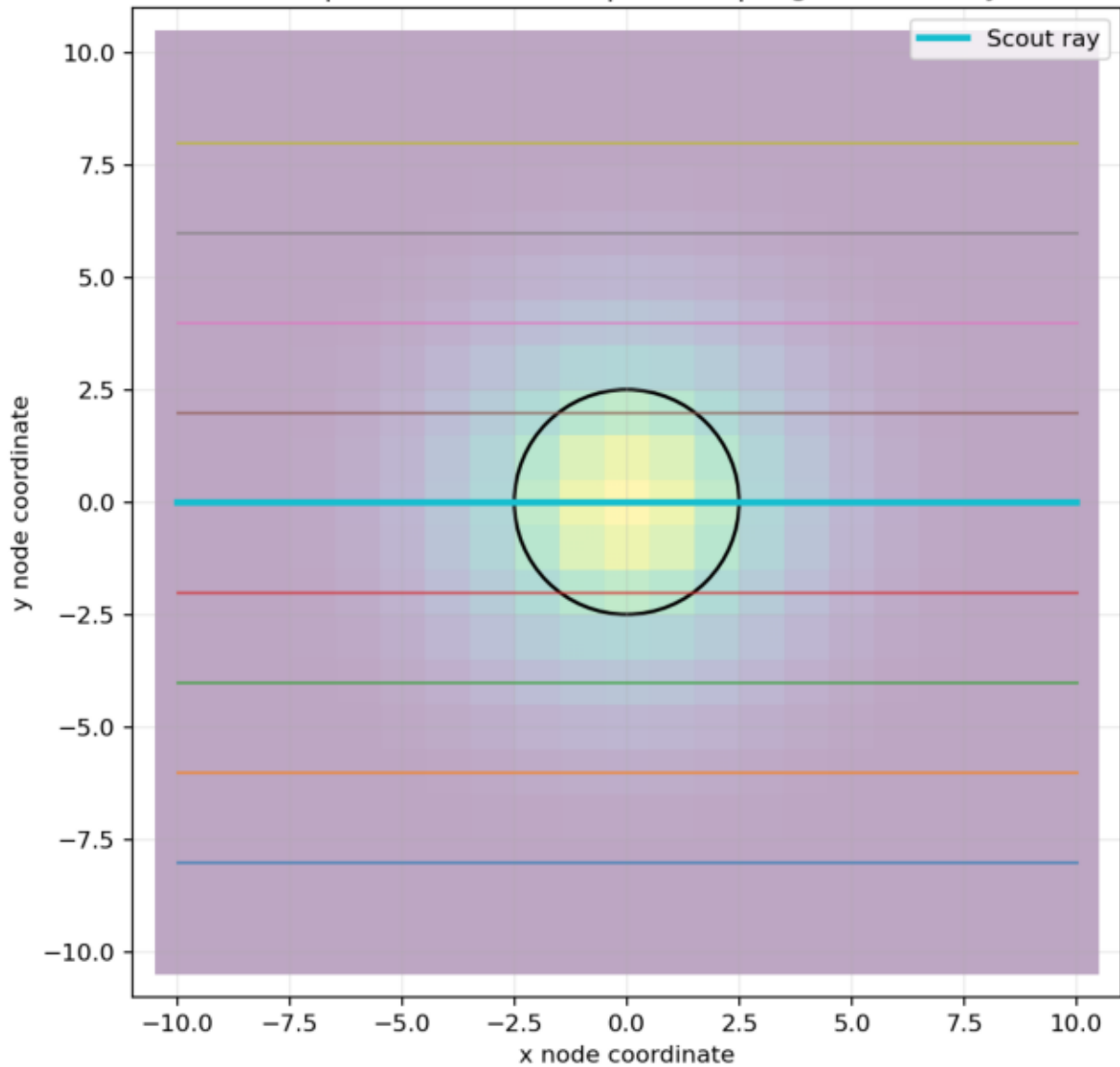
$\lambda=0$: scout cost=20.00, $L=0.00$, avg bundle delay=0.00

$\lambda=1$: scout cost=25.06, $L=5.06$, avg bundle delay=1.76

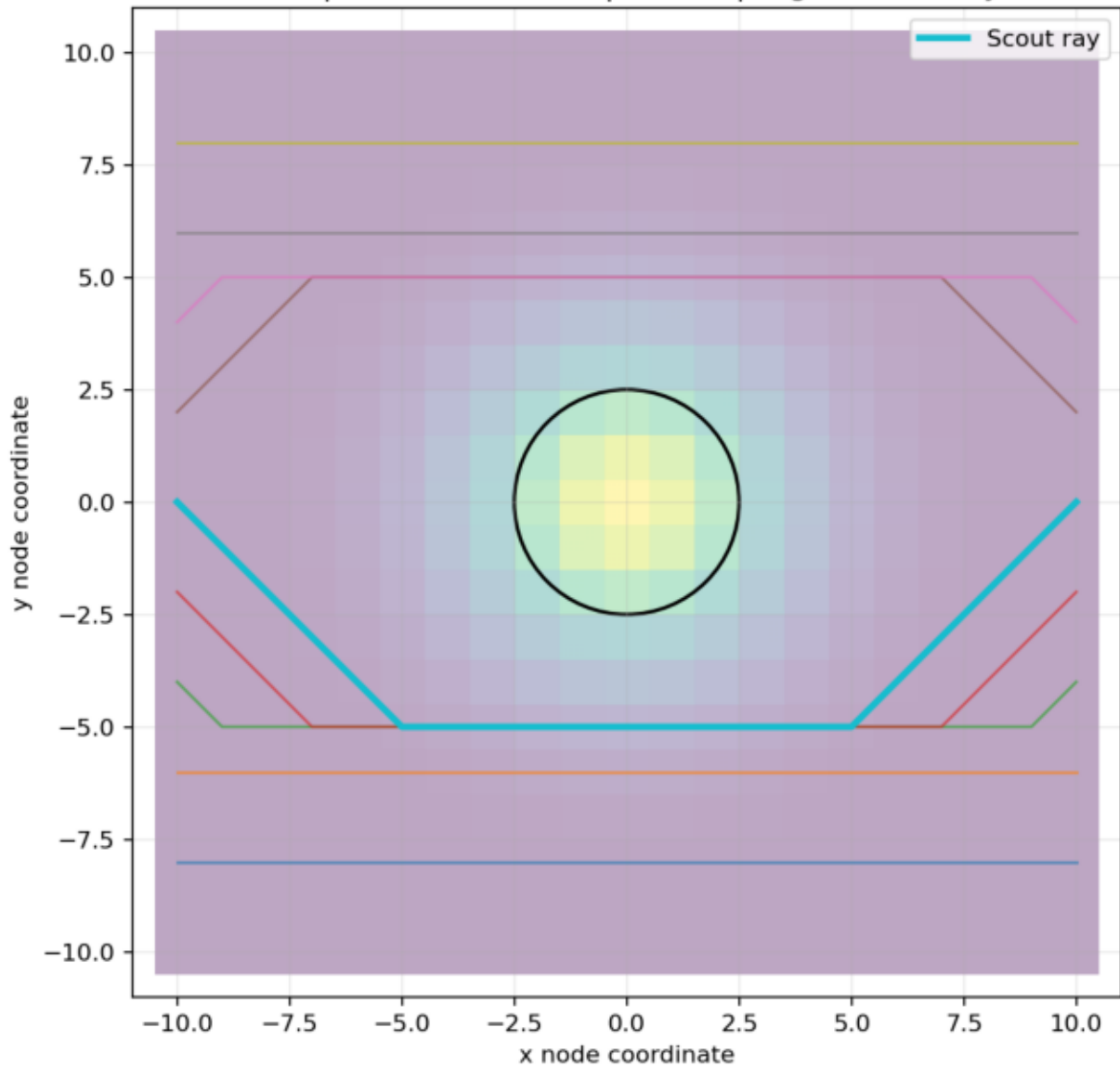
$\lambda=5$: scout cost=27.26, $L=7.26$, avg bundle delay=2.95

$\lambda=10$: scout cost=28.73, $L=8.73$, avg bundle delay=3.57

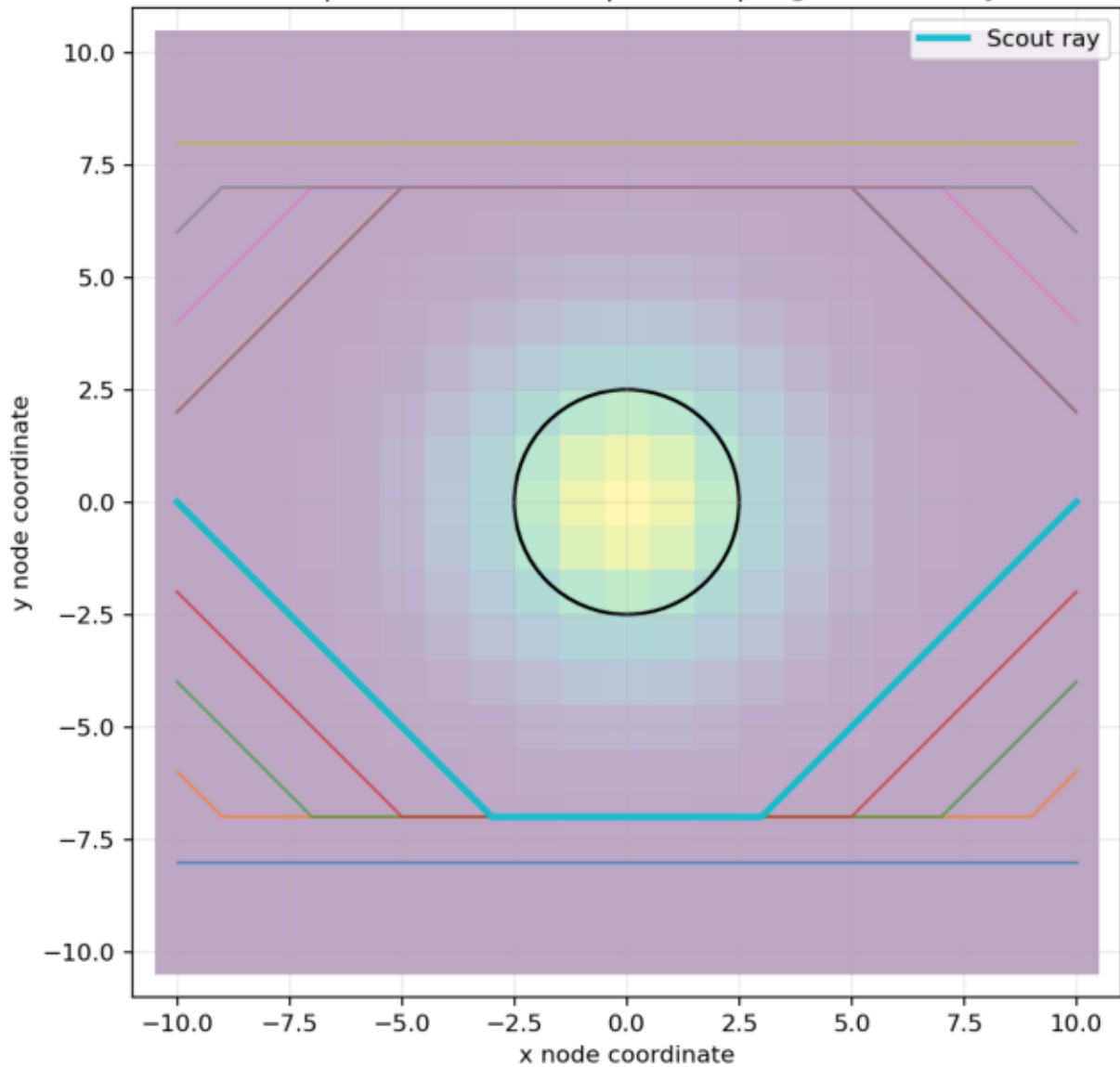
$\lambda=0$ | Scout cost=20.00 | $L=0.00$ | Avg bundle delay=0.00



$\lambda=1$ | Scout cost=25.06 | $L=5.06$ | Avg bundle delay=1.76



$\lambda=5$ | Scout cost=27.26 | $L=7.26$ | Avg bundle delay=2.95



$\lambda=10$ | Scout cost=28.73 | $L=8.73$ | Avg bundle delay=3.57

